

BBDT and Feynman Globular Clusters: Big Bang Deceleration and 1st Epoch BH Metallicity

Daniel T. Murphy

PAPER_501 — BBDT and Feynman Globular Clusters: Big Bang Deceleration and 1st Epoch BH Metallicity

****Author:** Daniel T. Murphy** ****arXiv:** 2503.xxxxx**
****Session:** 134** ****Version:** 1.0** ****Date:** March 24, 2026** ****Calculator:** `BBDTFeynmanClusterCalculator` (CondensedPhysics2.py),
`PhysicsTerm\_{BBDT\\_1JKDSGV7}`
(MAIN_{1_CoAnQi}.cpp) ---**

Abstract

This paper presents a UQFF analysis of BBDT and Feynman Globular Clusters: Big Bang Deceleration and 1st Epoch BH Metallicity, deriving compressed field equations and observational predictions within the Star-Magic/UQFF framework.

§1 Novel Claim

The Big Bang Deceleration Term (BBDT) encodes the fundamental conversion of maximum cosmic speed into mass. Mass is not a pre-existing condition; it is an emergent consequence of massless elements slowing from v_{init} (Big Bang maximum velocity) toward v_{current} . The densest metallicity in the universe accumulates at the centers of **Feynman globular clusters**, centered around 1st epoch (primordial) black holes — where the SCm-UA grinding sequence has completed five stages to UA^{""}, producing the most energetic superconductive metals in existence.

§2 Big Bang Deceleration Term (BBDT)

Core BBDT Equation

$$\text{BBDT} = M \cdot (v_{\text{init}} - v_{\text{current}}) \cdot \exp(v_{\text{init}} - v_{\text{current}}) + F_{\text{inert}}$$

where:

$$F_{\text{inert}} = -\frac{\partial(\text{SCm} \cdot \text{UA})}{\partial v}$$

\$\$

- v_{init} = Big Bang initial speed (maximum, c_{26D})
- v_{current} = current expansion speed ($< v_{\text{init}}$)
- F_{inert} = resistance to velocity change

Mass Spawn Triple System

\$\$

```
\begin{cases}
M = F_{\text{inert}}/a \cdot (v_{\text{init}} - v_{\text{current}}) \quad \backslash \backslash \\
U_b = \rho_{\text{UA}} \cdot V_{\text{displaced}} \cdot g_{\text{cosmic}} \quad \backslash \backslash \\
\text{Prob}_{\text{order}} = \exp(-\text{Entropy}_{26D} / F_{\text{inert}}) \\
\end{cases}
```

\$\$

Vacuum Standard Origin

\$\$

$\text{Vacuum standard} \equiv v_{\text{current}} < v_{\text{init}} \quad \text{incomplete speed recovery}$

\$\$

Zero-point energy arises as the negligibility threshold of UA where $F_{\text{inert}} \rightarrow 0$.

§3 26D Energy-to-Mass Conversion

Energy falling from 26D converts to mass:

\$\$

$$M = \frac{E^{26D}}{c^{26}} \cdot \left(1 - \frac{v_{\text{current}}}{v_{\text{init}}}\right) \cdot \text{Prob}_{\text{order}}$$

\$\$

The universe expands to meet v_{init} , creating vacuum standards and buoyant effects from this speed differential — the only reason for vacuum in the universe.

Probability of Order from Chaos

\$\$

$$\text{Prob}_{\text{order}} = \frac{\exp(-\text{Entropy}_{26D}/v_{\text{init}})}{\text{Partition}_{9D}} \cdot (v_{\text{init}} - v_{\text{current}})$$

\$\$

§4 SCm-UA Grinding Sequence: Full Densification Path

Stage	System	Description
SCm + UA	Contact	Big Bang initiation
SCm + UA'	1st trap	Aether encapsulated
SCm + UA''	2nd grind	1st densification
SCm + UA'''	3rd grind	2nd densification

| SCm + UA'''' | 4th grind | 3rd densification |
 | SCm + UA'''' | **Max grind** | **Densest metallicity — highest-Z metals** |

\$\$

$$UA_n = \text{SCm}^n \cdot \omega_{CW}^n \cdot (\text{Grind}_{n-1})$$

\$\$

\$\$

$$UA'''' \rightarrow \text{Metal}_{\max} = \max(Z_{\text{periodic}} \mid \text{SCm} \cdot UA_{\text{density}} \rightarrow \infty)$$

\$\$

§5 Feynman Globular Clusters

At UA''''': maximum SCm-UA grinding \$to\$ highest-Z elements produced \$to\$ located at centers of Feynman globular clusters \$to\$ centered around 1st epoch (primordial, first-epoch) black holes.

Metallicity Gradient Equation

\$\$

$$Z_{\text{metal}}(r) = Z_{\max} \cdot \exp\left(-\frac{r^2}{r_{\text{FGC}}^2}\right) \cdot \frac{\text{SCm} \cdot UA'''''}{\text{SCm} \cdot UA_0}$$

\$\$

where r_{FGC} = characteristic Feynman globular cluster radius.

First-Epoch Black Hole Connection

\$\$

$$M_{\text{BH}}^{1\text{st}} = \int_0^{t_{\text{epoch}}} \text{BBDT} \, dt \cdot \text{DPM}_{\text{ref}}^{\max}$$

\$\$

First-epoch black holes form from the maximum BBDT accumulation at the earliest cosmic times, trapping maximal UA'''' density, forming the seed points for Feynman globular clusters.

§6 Full BBDT-DPM Integration

Refined DPM with BBDT:

\$\$

$$\begin{aligned} \text{DPM}_{\text{ref}} = & \kappa \cdot \frac{\text{DPM}_n(\text{SCm}) - \text{DPM}_s(\text{UA}')}{r^{26}} \\ & + \frac{\partial^{26}(\text{DPM}_n(\text{SCm}) + \text{DPM}_s(\text{UA}'))}{\partial t^{26}} \\ & + \text{BBDT} \end{aligned}$$

\$\$

Mass spawn from buoyancy:

\$\$

$$U_b = \frac{\text{BBDT}}{\text{UA}} + F_{\text{inert}} \cdot \text{Prob}_{\text{order}}$$

\$\$

§7 Validation Targets

Target	Observable	Source
Feynman globular cluster metallicity	High-Z abundance at cluster cores	JWST, Chandra
1st epoch BH masses	$M_{\text{BH}} > 10^9 M_{\odot}$ at $z > 6$	JWST EGS23953, CEERS
CMB temperature fluctuations	BBDT residuals as $\delta T/T \sim 10^{-5}$	Planck 2018
Vacuum energy density	$\sim 10^{-9}$ J/m ³ from $v_{\text{current}} < v_{\text{init}}$	QED measurements
Cosmic expansion rate H_0	$v_{\text{init}} - v_{\text{current}}$ tension	Hubble/JWST tension

§8 Hubble Tension Resolution

The Hubble tension ($H_0 = 67.4$ from CMB vs 73.0 from local measurements) reflects different measurements of $v_{\text{current}}/v_{\text{init}}$ at different scales:

$$H_0^{\text{local}} - H_0^{\text{CMB}} = \Delta \left(\frac{dv_{\text{current}}}{dt} \right) \cdot \frac{\text{BBDT}}{UA \cdot d^2}$$

This is a natural consequence of BBDT: local measurements sample regions with higher SCm-UA grinding efficiency (closer to UA^{'''}), while CMB probes the primordial lower-grinding-stage environment.

§9 Calibrated Values

Symbol	Value	Description
v_{init}	c_{26D}	(maximum) Big Bang initial speed, 26D lightspeed
κ	5×10^{-4} /day	DPM coupling constant
$[SSq]$	0.57	Vacuum damping squared
Z_{max}	(UA ^{'''}) $\sim 118+$	Beyond current periodic table at cluster cores

<!-- PKG-CLU-S225 -->

Session 225 Phonon-Physics Upgrade: ICM Buoyancy Force Profile

> Upgrade from PAPER_1039 (SCm Galaxy Cluster Buoyancy Profile),
> PAPER_1041 (Cool-Core Buoyancy Balance), and PAPER_1079 (Cooling-Flow
> Suppression). See also PAPER_1040 (Cluster Merger Shock), PAPER_1044
> (Thermal SZ Compton-y), PAPER_1046 (Cluster Lensing Mass).

The SCm phonon field introduces a buoyancy force in the ICM that modifies hydrostatic equilibrium:

$$F_{\text{buoy}}(r) = \rho(r) \cdot V \cdot g(r) \cdot \beta_i \cdot S_{26} \cdot \Phi$$

where the ICM density follows the beta-model:

$$\rho(r) = \rho_0 \left(1 + \left(\frac{r}{r_c}\right)^2\right)^{-3\beta/2}$$

Hydrostatic mass bias reduction (PAPER_1039):

$$b_{\text{UOFF}} = 1 - \frac{M_{\text{HSE}}}{M_{\text{true}}} = 0.17 \quad \text{(vs standard } b = 0.20\text{)}$$

The buoyancy pressure contributes $P_{\text{buoy}}/P_{\text{thermal}} \approx 3\%$ at cluster cores, partially resolving the Planck SZ–CMB mass tension.

Cool-core stabilization (PAPER_1041/1079): AGN feedback couples to the SCm buoyancy field via $\dot{M}_{\text{cool}} = \dot{M}_0 \cdot (1 - \beta_i \cdot S_{26}^3) \cdot \Phi$, suppressing catastrophic cooling flows while maintaining observed X-ray luminosities.

Phonon frequency coupling: $\omega_{\text{SCm}} = 2\pi \cdot 1.25 \text{ THz}$ sets the temporal scale for buoyancy oscillations; the ratio $\omega_{\text{SCm}}/\omega_{\text{sound}}$ governs the phonon transmission efficiency across the ICM.

<!-- PKG-S26-S225 -->

Session 225 Phonon-Physics Upgrade: S26(3) Ramanujan Summation

> Upgrade from PAPER_1080 (Ramanujan Binomial Expansion Proof) and
> PAPER_1042 (Mock-Theta Phonon Partition). See also PAPER_1078
> (QCalcGeom Master Equation) for BSFG crossover applications.

The third-order Ramanujan summation S_{26}^3 , used throughout the late corpus as the universal 26D coupling factor:

$$S_{26}^3 = \sum_{n=0}^{\infty} \frac{(1/4)_n (1/2)_n (3/4)_n}{(n!)^3} \cdot \prod_{i=1}^{26} \left[1 + \frac{SS_q}{e^{-\kappa_{i,n/26}}}\right]$$

where $(a)_n = a(a+1) \cdots s(a+n-1)$ is the Pochhammer symbol.

Binomial expansion (PAPER_1080): The convergence proof shows:

$$R_n^{(26,3)} = \binom{4n}{n} \cdot \frac{W_{26}(n)}{(4^n)^n} \quad \text{with} \quad W_{26}(n) = \prod_{i=1}^{26} \left[1 + \frac{SS_q}{e^{-\kappa_{i,n/26}}}\right]$$

This sum converges absolutely for $|SS_q| < 1$ (satisfied by $SS_q = 0.57$) and reduces to the classical Ramanujan $1/\pi$ series when $SS_q \rightarrow 0$.

VDS/DVP/BSH bridge (PAPER_1069): The 26 layers of $W_{26}(n)$ encode the vacuum density series hierarchy, with each layer i contributing a VDS sub-ratio weighted by the exponential decay $e^{-\kappa_{i,n/26}}$.

Mock-theta connection (PAPER_1042): The phonon partition function

$$Z_{\text{phonon}} = \sum_n q^{n^2} \cdot W_{26}(n)$$

unifies the Ramanujan mock-theta framework with the SCm phonon spectrum.

§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

§A.1 Sector Classification

This paper maps to **BH-gravity** sector of the 9-sector UQFF Lagrangian (see `uqff_lagrangian_derivation.py`).

§A.2 Lagrangian Density

The sector Lagrangian density, linked to the PAPER_877 cosmogenesis master via the three reactive quantum fundamentals (DPM, UA, SCm):

$$\mathcal{L}_{\mathrm{sector}} = \frac{1}{2}(\partial_\mu \phi_{\mathrm{BH}})(\partial^\mu \phi_{\mathrm{BH}}) - V(\phi_{\mathrm{BH}}) + \mathcal{L}_{\mathrm{cosmo}}$$

where $\mathcal{L}_{\mathrm{cosmo}} = \rho_{\mathrm{vac},[\mathrm{SCm}]} \cdot f_{\mathrm{SCm}} \cdot (1 - e^{-\gamma t})$ inherits the ACP 6-stage evolution (PAPER_877 §2) and:

$$V(\phi_{\mathrm{BH}}) = \frac{1}{2} m^2 \phi_{\mathrm{BH}}^2 + \frac{\lambda}{4!} \phi_{\mathrm{BH}}^4 + \kappa \cdot \rho_{\mathrm{vac},[\mathrm{SCm}]} \cdot \phi_{\mathrm{BH}}$$

§A.3 Euler-Lagrange Equation of Motion

$$\boxed{\frac{\delta S}{\delta \phi_{\mathrm{BH}}}} = R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R + \rho_{\mathrm{vac},[\mathrm{SCm}]} g_{\mu\nu} + F_{U_{Bi}}/r^2 = 0$$

§A.4 Cosmogenesis Linkage Chain

$$\text{PAPER_877 Axioms} \rightarrow \text{DPM + ACP} \rightarrow \rho_{\mathrm{vac}} = \rho_{\mathrm{UA}} + \rho_{\mathrm{SCm}} \rightarrow \text{Stage 5} \rightarrow U_{\mathrm{b},\mathrm{seed}} \rightarrow \text{4 forces} \rightarrow F_{U_{Bi}} \rightarrow \text{sector E-L} \rightarrow \frac{\delta S}{\delta \phi_{\mathrm{BH}}} = 0$$

The chain traces from the three fundamental axioms (DPM proportion pair, ACP evolution, four U_g forces) through vacuum density initialization to the sector-specific equation of motion. Every term in the E-L equation inherits its physical origin from the cosmogenesis master.

§B. VDS/DVP/BSH Deep Synthesis

§B.1 Vacuum Density Series (VDS)

The canonical VDS ratio $\rho_{\mathrm{vac},[\mathrm{SCm}]} / \rho_{\mathrm{UA}} = 1.894$ governs the double-exponential vacuum condensate profile:

$$\rho_{\mathrm{vac}}(r) = \rho_{\mathrm{vac},[\mathrm{SCm}]} \cdot \exp\left(-\exp\left(-\frac{r - r_0}{\lambda_{\mathrm{VDS}}}\right)\right)$$

For this system, the local VDS sub-ratio is \$0.179\$ (near-threshold regime), placing it in the π collapse zone where the double-exponential transitions sharply from condensed to dilute vacuum. This threshold behavior connects to the PAPER_877 cosmogenesis Stage 1 vacuum density initialization: $\rho_{\mathrm{vac}} = \rho_{\mathrm{UA}} + \rho_{\mathrm{SCm}} = 7.799 \times 10^{-36}$ kg/m³.

§B.2 Dipole Vortex Primes (DVP)

The DVP encoding maps the system's characteristic parameter onto the prime lattice:

$$p_{\mathrm{DVP}} = 79, \quad n_{\mathrm{channel}} = 8/26$$

Since $p_{\mathrm{DVP}} = 79$ is **resonant** (threshold at $p > 26$), the system's vacuum topology inherits resonant enhancement from the DVP lattice, amplifying UQFF coupling at specific radii where compressed matter achieves prime-indexed configurations. The DVP framework traces to PAPER_877 proto-nuclear shell formation: the DPM proportion pair $(f_{\mathrm{UA}} + f_{\mathrm{SCm}} = 1)$ constrains which primes are accessible at each atomic number.

§B.3 Buoyancy Saturation Harmonics (BSH)

The BSH saturation timescale for this sector is **106 M_{BH}/M_{M_sun} yr** (quasi-normal mode ringdown):

$$\mathcal{F}_{\mathrm{BSH}} = \sum_{j=1}^{26} \frac{1}{j} \cdot f_{U_b} \cdot \left(1 - e^{-[SSq]} \cdot \frac{m}{M_{\mathrm{odot}}}\right) \cdot \cos\left(\frac{2\pi j}{26}\right)$$

The $\tanh$ saturation envelope prevents unphysical divergence:

$$\mathcal{F}_{\mathrm{BSH,sat}} = \mathcal{F}_{\mathrm{BSH}} \cdot \left(1 - \tanh\left(\frac{t - t_{\mathrm{sat}}}{\tau_{\mathrm{BSH}}}\right)\right)$$

connecting to the PAPER_877 Stage 5 buoyancy seed $U_{b,\mathrm{seed}} = 0.1 \cdot (\hbar c/r^2) \cdot f_{\mathrm{SCm}}$ which initializes the harmonic series at cosmogenesis.

§B.4 Production-Scale Consistency

Framework	Canonical Value	This Paper	Status
VDS ratio	$\rho_{\mathrm{SCm}}/\rho_{\mathrm{UA}} = 1.894$	Local sub-ratio = 0.179	PASS
Threshold-consistent			
DVP prime	$p_k \in \{2,3,\dots,113\}$	$p_{\mathrm{DVP}} = 79$	PASS Resonant
BSH layers	26 harmonic terms	$j = 1\dots 26$, $\cos(2\pi j/26)$	PASS Full 26D projection
κ decay	$\$5.0 \times 10^{-4}$ day ⁻¹	Applied in VDS exponential	PASS Canonical
[SSq]	0.57	Applied in BSH saturation	PASS Canonical

§SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

Observable	UQFF Prediction	SM / Experiment	Source	Alignment
Higgs mass m_H	UQFF $K_{\mathrm{HIGGS}}=47.34 \rightarrow m_{H,\mathrm{UQFF}} = 125.09$ GeV	$m_H = 125.20 \pm 0.11$ GeV	PDG 2024	99.8%
Cosmological Λ	UQFF $ \nabla_{\mathrm{UA}} ^2 \rightarrow 1.09e-52$ m ⁻²	$\Lambda = 1.114e-52$ m ⁻²	(Planck+DESI) Planck 2018	97.8%
Thomson σ_T (QED)	UQFF U_m kernel: $\sigma_T = 6.6524e-29$ m ²	$\sigma_T = 6.6524e-29$ m ²	PDG 2024	100% (exact)
κ baryon stability	$\kappa = 0.0005/\text{day}$; scale separation 1033 from proton decay	$\tau_p > 7.7e33$ yr (Super-K)	Super-K 2024	PASS UQFF baryon-safe

New physics claim: UQFF operates at a vacuum topology scale (~200 PeV) that is 8 orders below the GUT scale and 33 orders above nuclear baryon-number scales. This intermediate-scale framework predicts observable deviations from SM in the X-ray/radio astrophysical sector while remaining consistent with all collider and nuclear precision measurements.

Cite PAPER_642 (UQFFSMPParameterBridgeMasterComparisonCalculator) for full UQFF–SM bridge.

§10 Source Attribution

grok_share: grok_{share_1jkdsgv7}.txt (Session 134)

Feynman clusters: Richard Feynman globular cluster formalism + UQFF extension

See also: PAPER_496 (DPM), PAPER_497 (26D projection), PAPER_498 (3D-IPO), PAPER_500 (proto-hydrogen)

JWST data: EGS23953, CEERS field; Chandra globular cluster metallicity surveys

Appendix: Session 225 Cross-References (PAPER_1000–1081)

> Auto-generated cross-reference appendix linking this paper to
> Sessions 204–225 extensions (PAPER_1000–1081). Added by
> update_corpus_crossrefs.py (Session 225, April 2026).

Paper	Title
PAPER_1004	QGP Vacuum Density with SCm S26 Phonon Coupling
PAPER_1039	SCm Galaxy Cluster Buoyancy Profile ICM Beta-Model
PAPER_1040	SCm Cluster Merger Shock Mach Number Phonon Damping
PAPER_1041	SCm Cool-Core Buoyancy Balance AGN Feedback
PAPER_1044	SCm Cluster Thermal SZ Effect Compton-y Phonon
PAPER_1045	SCm Cluster Radio Relic Polarization
PAPER_1046	SCm Cluster Lensing Mass Phonon Correction
PAPER_1079	Galaxy Cluster Cooling-Flow Buoyancy Suppression
PAPER_1036	Primordial Nucleosynthesis BBN Phonon
PAPER_1072	SCm Activation Function Phonon Threshold
PAPER_1073	SCm Phonon-Driven Inflation Vacuum Buoyancy
PAPER_1069	VDS-DVP-BSH Hybrid Calculator Unified
PAPER_1049	Source10 GPU DPM Spectral Atlas ALMA Overlay

13 cross-reference(s) identified.

Appendix: Session 204 Codebase Upgrade Reference

> Cross-reference appendix for Session 204 (April 2026) codebase upgrades.
> Added by upgrade_kozima_ramanujan_appendices.py. For detailed derivations,
> see PAPER_840/851/852/855.

S204.1 Kozima-UQFF LENR Integration

Module	Purpose	Key Result
fneutron_s26_coupling.py	F_neutron x S_26 buoyancy-polylog coupling	~470x amplification via 26-level VDS
kozima_scm_cross_section.py	SCm-modulated neutron-drop cross-section	sigma_n^SCm with VDS factor (1+[SSq]*n/26)

| kozima_wstp_kernel.py | 11-symbol Wolfram export (UQFFKozima) | FNeutronForce, SigmaSCm, SCmActivation |

Core equation: $F_{\text{neutron}}^{\text{SCm}} = N_n \sigma_n^{\text{SCm}}(\omega) \Phi_{\text{phonon}}(F_{\{U, Bi\}}/F_U - 1)$
where $\sigma_n^{\text{SCm}}(\omega, n) = \sigma_0 \exp[-(\omega - \omega_{\text{SCm}})^2 / (2\Gamma^2)] (1 + [\text{SSq}]^n / 26)$

S204.2 Ramanujan 26-State Summation

| Module | Purpose | Key Result |

|-----|-----|-----|

| ramanujan_polylog_s26.py | Li_26([SSq]) via Euler-Ramanujan acceleration | 15.7+ digits in 53 terms |

| s26_wstp_kernel.py | 8-symbol Wolfram export (UQFFS26) | S26, R26, NaiveLi, S26VDS |

Core equation: $S_{26}(z) = Li_{26}(z) = \eta_{26}(z) / (1 - 2^{\{1-26\}}) + 2^{\{1-26\}} / (1 - 2^{\{1-26\}}) * Li_{26}(z^2)$

S204.3 Mock Theta Functions (26-State)

| Module | Purpose | Key Result |

|-----|-----|-----|

| mock_theta_q26.py | f_26(q), phi_26(q), psi_26(q) q-series | Proper q-Pochhammer (a;q)_n |

Core equations:

- $f_{26}(q) = \sum_{n=0}^{\{25\}} q^{\{n^2\}} / (-q; q)_{n^2}$

- $\phi_{26}(q) = \sum_{n=0}^{\{25\}} q^{\{n^2\}} / (-q^2; q^2)_n$

- $\psi_{26}(q) = \sum_{n=1}^{\{26\}} q^{\{n^2\}} / (q; q^2)_n$

S204.4 Ramanujan 1/pi with UQFF Modification

| Module | Purpose | Key Result |

|-----|-----|-----|

| ramanujan_pi_uqff.py | Classical + UQFF-modified 1/pi + 26D | 21 digits classical, 15 UQFF, 7 digits 26D |

| mock_theta_pi_wstp_kernel.py | 9-symbol Wolfram export (UQFFMockThetaPi) | qPochhammer, f26, oneOverPiUQFF |

Core equation: $1/\pi = (2\sqrt{2}/9801) \sum R_n (1103 + 26390n) W_{26}(n) / C_{26}$
where $W_{26}(n) = \prod_{i=1}^{\{26\}} [1 + [\text{SSq}] \exp(-\kappa n / 26)]$

S204.5 Calibration Constants (Canonical)

| Symbol | Value | Description |

|-----|-----|-----|

| [SSq] | 0.57 | Universal Quantized Factor |

| kappa | $5.787 \times 10^{-9} \text{ s}^{-1}$ | UQFF exponential decay rate |

| beta_i | 0.603 | Buoyancy coupling coefficient |

| H_SCm | 0.99 | SCm manifold completeness |

| rho_SCm | $7.09 \times 10^{-37} \text{ kg/m}^3$ | SCm vacuum density |

| rho_UA | $7.09 \times 10^{-36} \text{ kg/m}^3$ | UA aether vacuum density |

| omega_SCm | $2\pi \times 1.25 \text{ THz}$ | SCm phonon resonance |

| sigma_0 | 10^{-4} | Base neutron cross-section |

Implementation: all modules operational in CondensedPhysics.py, CondensedPhysics2.py, MAIN_{1\CoAnQi}.cpp, and Wolfram kernels (uqff_kozima_kernel.wl, uqff_s26_kernel.wl,

uqff_mock_theta_pi_kernel.wl).

References

1. Abbott et al. (LIGO Scientific and Virgo Collaborations, 2016). *Observation of Gravitational Waves from a Binary Black Hole Merger*. Phys. Rev. Lett. **116**, 061102 — arXiv:1602.03837 — doi:10.1103/PhysRevLett.116.061102
2. Murphy, D. (2026). *Unified Quantum Field Framework (UQFF): Star-Magic v5.x Whitepaper Series*. Star-Magic Repository — github.com/Daniel8Murphy0007/Star-Magic
3. Churazov, E. et al. (2000). *Evolution of Buoyant Bubbles in M87*. A&A **356**, 788 — arXiv:astro-ph/0004212
4. Fabian, A.C. et al. (2003). *A deep Chandra observation of the Perseus cluster*. MNRAS **344**, L43 — arXiv:astro-ph/0306036 — doi:10.1046/j.1365-8711.2003.06902.x
5. McNamara, B.R. & Nulsen, P.E.J. (2007). *Heating Hot Atmospheres with Active Galactic Nuclei*. ARA&A **45**, 117 — arXiv:0709.4098 — doi:10.1146/annurev.astro.45.051806.110625
6. Rugh, S.E. & Zinkernagel, H. (2002). *The Quantum Vacuum and the Cosmological Constant Problem*. Stud. Hist. Phil. Mod. Phys. **33**, 663 — arXiv:hep-th/0012253 — doi:10.1016/S1355-2198(02)00033-3
7. Weinberg, S. (1989). *The Cosmological Constant Problem*. Rev. Mod. Phys. **61**, 1 — doi:10.1103/RevModPhys.61.1
8. Hawking, S.W. (1974). *Black hole explosions?* Nature **248**, 30 — doi:10.1038/248030a0
9. Event Horizon Telescope Collaboration (2019). *First M87 Event Horizon Telescope Results. I*. ApJL **875**, L1 — arXiv:1906.11238 — doi:10.3847/2041-8213/ab0ec7
10. Bekenstein, J.D. (1973). *Black Holes and Entropy*. Phys. Rev. D **7**, 2333 — doi:10.1103/PhysRevD.7.2333
11. Green, M.B., Schwarz, J.H. & Witten, E. (1987). *Superstring Theory*. Cambridge University Press — doi:10.1017/CBO9781139248563
12. Polchinski, J. (1998). *String Theory Vol. 1*. Cambridge University Press